



GUIDE TO BUYING WINDOWS

UNDERSTAND PRICING, PRODUCTS, AND MORE

Table of Contents

Welcome Message	3
Budgeting for Windows	4
Window Selections Worksheet	7
Installation Type	8
Materials	11
Energy Efficiency	13
Window Style	18
Window Installation	29
Get in Touch!	34

Welcome Message

Buying windows is a confusing process, and most homeowners only buy windows once in their lifetime. You're not alone if you have no idea what to look for in a window or a window installation company! The average homeowner has never had to think about what is a fair price for windows and window installation.

We created this guide to help overwhelmed homeowners plan their budget and choose windows with confidence.

Our company, Window Works Co., is dedicated to helping homeowners in the Raleigh, NC area find the windows and doors they'll love.

Loving your windows means not feeling confused, overwhelmed, pressured, or ripped off at any point in the process.

This book will guide you through what you need to know to help you find the exact windows you're looking for. Whether you are searching for the highest energy efficiency, the most durable materials, or the latest trends in style and color, we hope that this guide will put you at ease, and lead you to a window that is perfect for your home and your family.



Jeff and Justin Reif

Budgeting for Windows

Buying windows is a lot like buying a house. How much you spend will depend on a variety of factors. These include the type of replacement you choose, your windows' energy efficiency, the materials your windows are made from, the style of window you choose, and other choices you make related to the installation process.

In this guide, we'll explain each of those factors, so that you understand your options and can prioritize what's most important to you. That might be budget, durability, ease of maintenance, aesthetic, or some combination of these.

A good window that is well installed can easily last 30 years or more. We encourage homeowners to view their windows, like their home, as an investment.

 Throughout this guide, we will highlight decisions you can make in the buying process that represent **long term investment**. While these options may have more upfront cost, they will ultimately save you money, time, and hassle. Look for the dollar sign to spot these tips.

Budgeting for Windows

Our customer's costs for windows and installation tends to **range from \$7,000 to \$100,000**. The lower end of this range represents installations that are pocket/insert replacements of fewer windows. The upper end represents projects with more windows, made from more durable materials in a full-frame replacement, with greater energy efficiency, and more customization of size, shape, and color.

The average (mean) cost of a window replacement project done by Window Works Co. since 2021 is \$22,745.

It is unlikely that a project replacing a home's windows can be accomplished for less than \$10,000.00.

However, a project with many windows, high energy efficiency demands, unique customization options, and top-of-the-line materials can certainly cost more than \$50,000.00.



Replacing windows is one of the most significant investments you will make into your biggest investment; your home. [On average, replacing windows instantly adds \\$13,766 - \\$14,912 in value to your home.](#) This means that customers who spend close to our average on a replacement project will make back almost all of their investment instantly in terms of their home's value.

For this reason, many homeowners **use financing to minimize their upfront cost, viewing this as an investment in their home's value.**

[Learn More About Our Financing Options](#)

Budgeting for Windows

Splitting A Window Replacement Project into Phases to Save \$?

Many homeowners think that breaking their project into smaller jobs is a good way to save money on window replacement.



As a rule, it will cost more money overall to split your window replacement into multiple phases. Here are several reasons why:

Window companies typically raise their prices annually. This **price increase occurs in January for most major brands**. This means that you will likely pay more per window for any later phase of your project.

A whole house installation job that could be completed in five days would likely take six or seven days if split into phases. That's because the installation crew would have to duplicate some work, like set up, break down, materials delivery, etc. Plus, you'd be paying some fees twice (like the dump fees required to dispose of your original windows). **Installation costs are always higher on a phased job than they would be if the job were completed all at once.**

Finally, it is rare, but possible, that a window company might discontinue or change the window you start your project with. This could make it tricky, and probably more costly, to customize a different window to look like your original selection.

If for some reason you still want to split your project into phases, we recommend you do this by elevation. [Read this article to learn more about what that means and how to do it.](#)

Window Selections

Use this quick, printable checklist to keep track of your decisions

Installation Type

- ☐ Full Frame ☐ Pocket/Insert

Material

- ☐ Fiberglass/Composite ☐ Vinyl ☐ Wood
☐ Aluminum-Clad ☐ Vinyl-Clad

Energy Efficiency

- ☐ ENERGY STAR ☐ No ENERGY STAR requirement

Window Type (May be Multiple)

- ☐ Double-Hung ☐ Single-Hung
☐ Casement ☐ Picture
☐ Sliding/Gliding ☐ Specialty (shapes, privacy glass, etc.)

Notes:

Color/Finish

Exterior Color:

Interior Color:

Grilles/Glass Options

Notes:

Installation Type

Before you make any other decisions, you should decide what type of window installation you want. There are two types to choose from.

Full Frame Replacement

During a full frame replacement, the entire existing window is removed and replaced with an all new window.

First, installers remove the existing window and the trim on both the inside and outside of the window. This creates a rough opening down to the wall studs. Finally, a completely new window is installed.

This is a costlier and more time-consuming option than a pocket/insert replacement is.

Pocket/Insert Replacement

In an insert window replacement, installers remove the sash and hardware of your existing window. They then install a new window within the existing window frame. The resulting window will have less glass area than the original window had.

This type of replacement is ideal for medium to large sized windows with structurally sound frames. Put another way, windows with no energy efficiency or moisture/leakage concerns.

Insert window replacement is not suited for older windows where homeowners suspect rot, or where warping or other use issues are noted.

Installation Type

Why Start Here?

Choosing your installation type first saves you time in your research process. It will also help you establish your budget, since installation type is a major factor in determining your window replacement budget.

Here is an example of how choosing installation type first can save you time and hassle during the research process.

Suppose you are replacing a vinyl window. In this case, you must do a full-frame replacement. That's because vinyl windows can't be pocketed inside an existing vinyl frame.

Knowing this, you can anticipate needing to account for a full-frame replacement in your budget.

Installation Type

This quick comparison chart can help you decide which installation type is right for you.

FEATURES	FULL FRAME	POCKET INSERT
ENTIRE WINDOW IS REPLACED	✓	
ONLY SASH AND HARDWARE ARE REPLACED		✓
CAN CHANGE WINDOW SIZE	✓	
RESULTING GLASS AREA IS SMALLER THAN ORIGINAL WINDOW		✓
CAN ONLY BE DONE ON STRUCTURALLY SOUND WINDOWS		✓
CAN BE DONE ON A VINYL WINDOW	✓	
ALLOWS FOR CHANGE OF WINDOW STYLE	✓	
LESS EXPENSIVE		✓
LESS TIME CONSUMING		✓

Materials

Next, you will decide what you want your windows made from. Here are your options. We have organized these from what tends to be most expensive to what tends to be least expensive.

Fiberglass/Composite	Extremely durable, resistant to rot, warp, and damage. Wide variety of colors and finishes available.
-----------------------------	---

Vinyl-Clad	Vinyl-clad windows have a wood interior and a vinyl exterior. They are durable, and a good choice for homeowner's who like the look of wood without the high maintenance costs.
-------------------	---

Aluminum-Clad	Extremely durable, resistant to rot, warp, and damage. Wide variety of colors and finishes available.
----------------------	---

Vinyl	Good durability (very dependent on the specific manufacturer's process). Somewhat resistant to rot, warp, and damage. Several color options and finishes available.
--------------	---

Wood	Least durable. Prone to rot, warp, and damage. Most likely to experience seal failure. Few pre-finished options, mostly raw or primed wood.
-------------	---

Materials

FEATURES	FIBERGLASS/ ALUMINUM- CLAD	VINYL	VINYL-CLAD	WOOD
UPFRONT COST	HIGH	MODERATE	HIGH- MODERATE	LOW
DURABILITY/ WEATHER RESISTANCE	EXCELLENT	MODERATE	HIGH	POOR
DEGREE OF MAINTENANCE	LOW	MODERATE	LOW	HIGH
LONG TERM VALUE	HIGH	MODERATE	HIGH	POOR
TYPICAL WARRANTY TERM (VARIES BY MANUFACTURER)	20 YEARS	LIFETIME	20 YEARS	1 YEAR



North Carolina's heat and humidity can cause windows to expand and contract. Over time, this can lead to seal failure, warp, and moisture damage. Choosing a durable window material will save you repair and replacement costs over time.

Fiberglass, fiberglass composite, aluminum-clad, vinyl-clad, and vinyl windows are the most durable windows. They are generally unaffected by even extreme temperature swings and high humidity the way that wood and vinyl windows are.

Warranties on these windows are lengthier than warranties on wood windows, because of the longevity you can expect from them.

Energy Efficiency

There are two main reasons to consider energy efficiency when replacing your windows.

Cost

The average American household spends \$1,500 – \$2,500 on energy costs annually. Around 45% of that cost is for heating and cooling. The energy efficiency of your windows is a primary factor in your home's heating and cooling costs.

Code

Depending on the extent of your window replacement project, you may need to adhere to local building codes.

What Makes a Window Energy Efficient?

What makes a window energy efficient is the degree to which the frame material and the glass prevent heated or cooled air from escaping your home. This may be accomplished through:

- Multiple panes of glass can help improve insulation
- Insulating gases: Gases like argon or krypton can be filled between the panes to help prevent heat transfer
- Low-emissivity (Low-E) coating that help reduce heat transfer
- Window frames made from materials that resist heat and cold transfer

Energy Efficiency

There are 3 steps to understanding a window's energy efficiency. We'll walk you through each on the following pages.

- Know what NFRC ratings are
- Find your ENERGY STAR climate zone
- Compare the NFRC ratings on the window you're considering to the requirements for your ENERGY STAR zone.

Want to Know More?

For most homeowners, it's enough to know that NFRC ratings exist and are used to determine a window's energy efficiency based on their climate zone (this is explained on the following pages).

If you want a more in-depth understanding of what each rating means, and how certain window features affect ratings, read this article:

[**The Complete Guide to Buying Energy Efficient Windows.**](#)



Energy Efficiency

NFRC Ratings

[The National Fenestration Rating Council \(NFRC\)](#) is a non-profit organization in the United States. The NFRC maintains an energy efficiency certification and labeling program for windows and skylights.

The NFRC provide performance standards for manufacturers' ratings for these products in five categories:

- U-Factor
- Solar Heat Gain Coefficient
- Visible Transmittance
- Air Leakage
- Condensation Resistance



A sample NFRC energy performance rating sticker. It features the NFRC logo and a star icon. The text on the sticker includes: 'World's Best Window Co.', 'Series "2000"', 'Casement', 'Vinyl Clad Wood Frame', 'Double Glazing, Argon Fill, Low E', 'XVZ-X-1-00001-00001'. The 'ENERGY PERFORMANCE RATINGS' section shows a U-Factor (U.S. I/F) of 0.35 and a Solar Heat Gain Coefficient of 0.32. The 'ADDITIONAL PERFORMANCE RATINGS' section shows Visible Transmittance of 0.51 and Air Leakage (U.S. I/F) of ≤0.3. Condensation Resistance is listed as 51. A disclaimer at the bottom states: 'Manufacturers represent that these ratings conform to applicable NFRC procedures for determining energy product performance. NFRC ratings are determined by third party measurement agencies and do not constitute any NFRC endorsement or approval of any product or claim. NFRC does not warrant the validity of any product for any specific use. Contact manufacturers' websites for other product performance information. www.nfrc.org'.

ENERGY PERFORMANCE RATINGS	
U-Factor (U.S. I/F)	Solar Heat Gain Coefficient
0.35	0.32
ADDITIONAL PERFORMANCE RATINGS	
Visible Transmittance	Air Leakage (U.S. I/F)
0.51	≤0.3
Condensation Resistance	
51	—

Sample
NFRC
Sticker

The NFRC puts a rating sticker like the one above on every window they have evaluated for efficiency.

These ratings serve several purposes. They help architects, builders, and homeowners compare products in regards to their energy efficiency. They also help builders and home designers ensure that the products they are considering meet local building codes.

Energy Efficiency

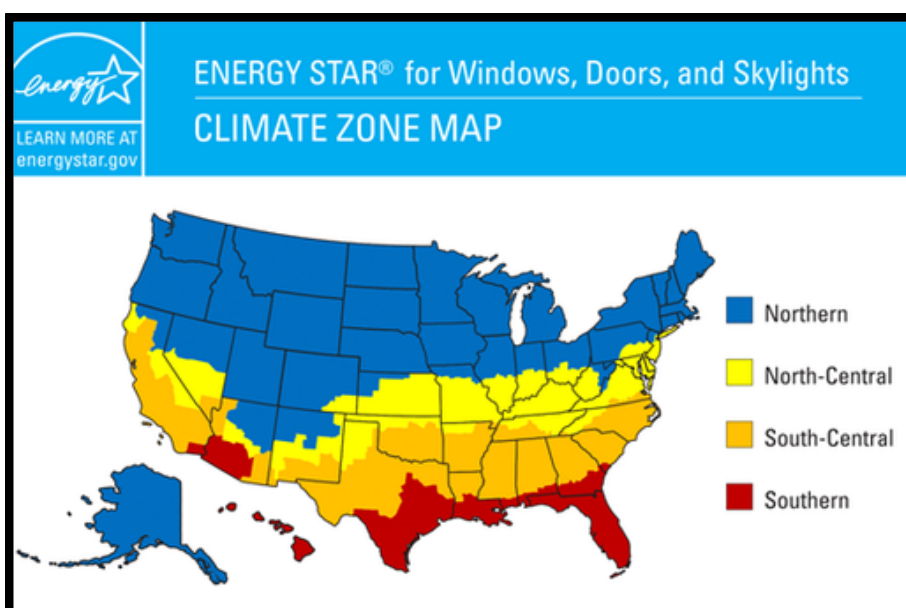
ENERGY STAR

ENERGY STAR is a program run by the U.S. Department of Energy and the U.S. Environmental Protection Agency. The program promotes energy efficiency by labeling a wide range of products as meeting ENERGY STAR requirements.

A product having an ENERGY STAR certification means that it meets a minimum standard of energy efficiency. [An ENERGY STAR certification indicates that a window:](#)

- is manufactured by an ENERGY STAR partner, and
- is independently tested, certified, and verified by the NFRC, and
- has NFRC ratings that meet strict energy efficiency guidelines set by the US Environmental Protection Agency (EPA).

The EPA divides the United States into four climate zones.



Raleigh, NC is currently in the **South-Central Climate Zone.**

Prior to October, 2023, Raleigh was in the North-Central Climate Zone.

Energy Efficiency

ENERGY STAR

In order to meet ENERGY STAR criteria for the South-Central climate zone, a window's NFRC ratings must meet the following criteria.

Windows

Climate Zone	U-Factor ¹	SHGC ²	
Northern	≤ 0.22	≥ 0.17	Prescriptive
	$= 0.23$	≥ 0.35	Equivalent Energy Performance
	$= 0.24$		
	$= 0.25$	≥ 0.40	
	$= 0.26$		
North-Central	≤ 0.25	≤ 0.40	
South-Central	≤ 0.28	≤ 0.23	
Southern	≤ 0.32	≤ 0.23	

Air Leakage for windows ≤ 0.3 cfm/ft²

¹ Btu/h ft²·°F

² Solar Heat Gain Coefficient

Armed with this information, you can evaluate the energy efficiency of the windows you're considering purchasing. You can be sure that they will not only look great, but that they will also help you cut back on your home's energy expenditure.

Window Style

Now it's time to make decisions about the style of window you want. There are several components of style to consider. This page offers an overview. We'll provide more detail about each style consideration on the pages that follow.

Window Type

Double hung, single hung, casement, gliding, and picture are the most common window types.

Color and Finish

You will have choices for both exterior and interior color, as well as finishing options.

Grilles

Depending on your home's architectural style, you may want your new windows to have grilles. You will have to choose a pattern and an application style.

Glass

You may want privacy glass in certain rooms like bathrooms, or you may want a sunny side of your home to have glass that is more energy efficient.

Hardware

Depending on the window you choose, you may have to decide on colors or safety features for locks and other hardware.

Window Style: Window Type

Here are some of the most common window types.

Double Hung

Also known as an 'Up and Down' window, double hung windows have two separate sashes, each of which can operate independently.



Pros

- Usually feature tilt-in option for easier cleaning
- Safer for kids/pets (can open top sash)
- More customization options (vs. single-hung)
- Generally a better built product than single-hung

Cons

- More expensive than single-hung windows

Window Style: Window Type

Single Hung

On a single-hung window, only the lower sash is operable. This means that you can move the bottom sash up to open the window. But the upper sash is stationary, and you cannot move it.



Pros

Less expensive than double-hung windows

Cons

More difficult to clean

Generally limited customization options

Typically a 'bare-bones', basic grade option

Still Can't Decide?

[If you want a more in-depth comparison of double-hung and single-hung windows, read this article.](#)

Window Style: Window Type

Picture Window

A picture window does not open. Picture windows are ideal for rooms with great views and limited ventilation needs.

Modern and contemporary architectural styles usually make use of picture windows.

Windows in unique shapes (circles, triangles, trapezoids, etc.) are typically picture windows.



Pros

Can be made very large

Cons

Exterior must be cleaned from outside home
Don't offer ventilation

Window Style: Window Type

Casement Window

Casement windows look like picture windows, but they can be cranked open to allow limited ventilation



Pros

Allow ventilation

Cons

More difficult to clean

Opening may interfere with outside landscaping if they open outward

Window Style: Window Type

Gliding/ Sliding Window

Gliding or sliding windows open by sliding horizontally. Most offer an interior tilt cleaning option.



Pros

- Allow ventilation
- Easy to clean
- Look like casement windows but are generally less expensive

Cons

- Gliding track may require minimal maintenance to avoid debris buildup

Window Style: Color and Finish

Next, you should decide what color you want your windows to be. You will need to choose an interior and exterior color.

At this point, having a rough ideal of the color you want is sufficient. Until you choose a window brand, you will not know exactly what color options you have.

However, suppose you have already decided that you want a pocket/insert replacement with a vinyl window. You want the exterior of your window to be black, and the interior to be white.

This will help you narrow down window brand options tremendously. That's because many vinyl window manufacturers don't offer a black exterior vinyl window.

One brand that does, that we at Window Works Co. are proud to carry, is [ProVia](#). So, if you told us you want a black vinyl window, we would know immediately to direct you to a ProVia window, since they can meet your exact needs.

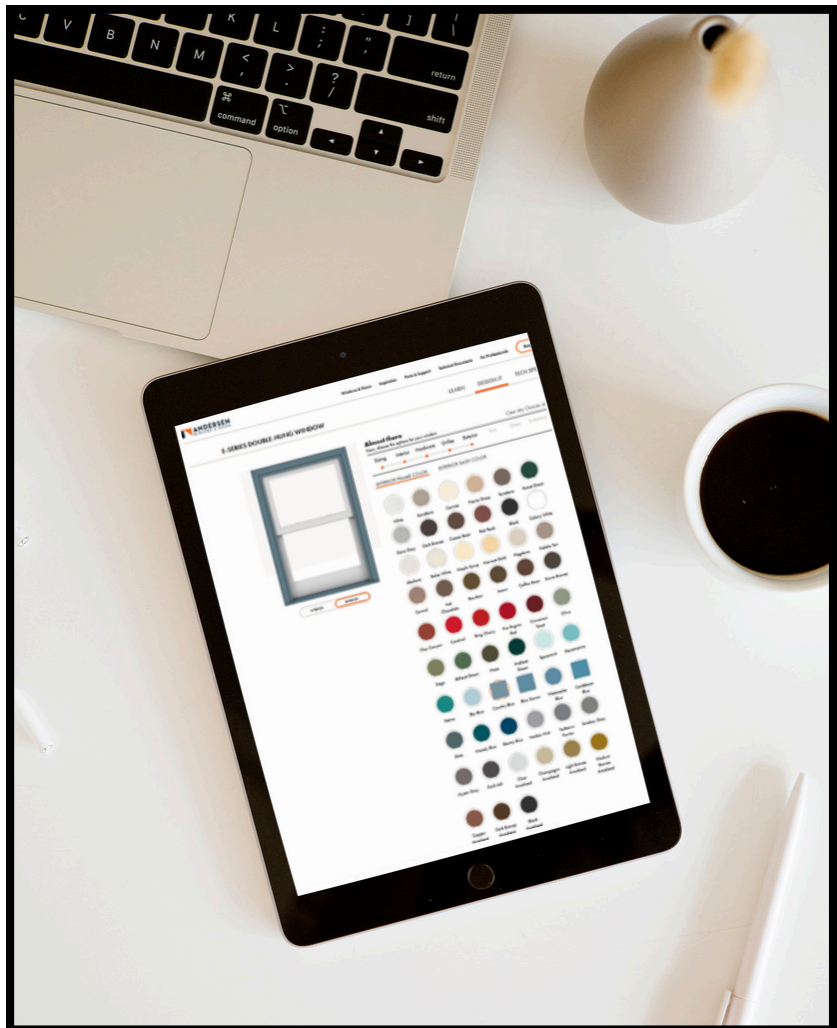


Window Style: Color and Finish

Once you've determined which window brand you'd like to go with, you can review the specific color options available in the windows you're considering.

Color options typically vary based on other features you've already decided on, such as the material and window type.

Most leading window brands offer color selector tools on their websites.



Window Style: Grille Options

At this point, you should decide if you want grilles (also called grids) for your windows. These are the most common grille options.

No Grilles

It's certainly an option not to have grilles. Modern and Contemporary style homes often include large picture windows, which may look better without grilles.

Grilles Between the Glass

Some windows offer grilles between the glass. Here, the grille pattern is inlaid between the layers of window glass. This reduces the risk of grilles breaking and makes cleaning much easier.

Removable Interior Grilles

These grilles attach to the inside of a window. They can be easily removed for easy cleaning or to change the look of a window

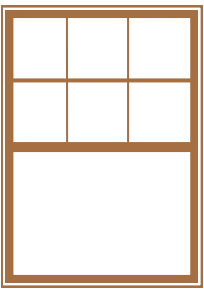
Simulated Divided Lite

The earliest windows were built with many small panes of glass held together by grids of wood, and were called 'True Divided Lites'.

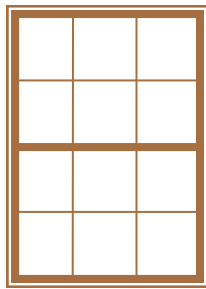
This grille style mimics this style, but without using individual pieces of glass. The grille is welded to the window frame, and can't be removed.

Window Style: Grille Patterns

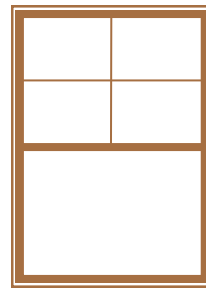
There are endless grille pattern options available. These are some of the most common.



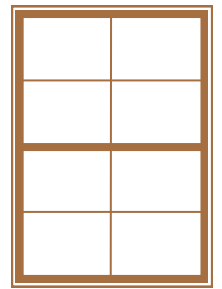
Six over One



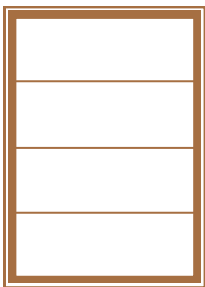
Six over Six



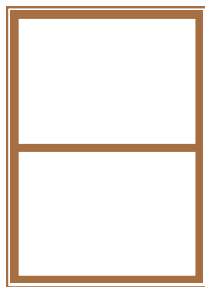
Four over One



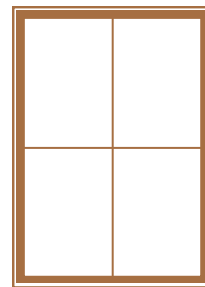
Four over Four



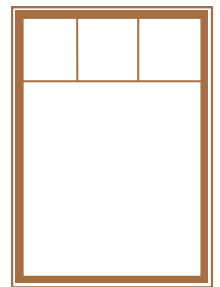
Equal Lite



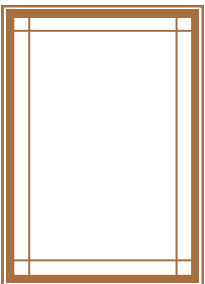
Simulated
Double Hung



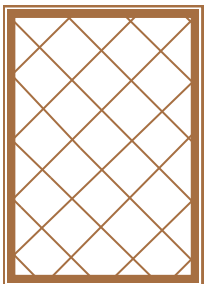
Two by Two



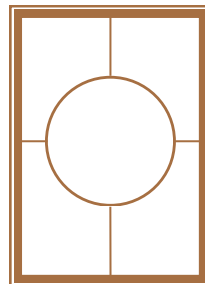
Short
Fractional



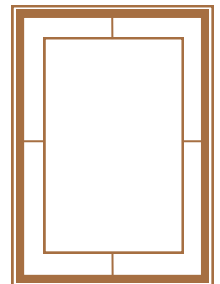
9-Lite Prairie



Diamond



Custom



Custom

Window Style: Glass, Hardware, and Screens

Now it is time to make a note of any finishing preferences you have. These might be related to glass, hardware, or screens.

For instance, you may want obscured or privacy glass for certain rooms, like bathrooms.

Or, you may want an added degree of energy efficiency for rooms that receive a lot of sunlight. You can accomplish this by choosing low-E glass for just that elevation (or side of your home).

Similarly, you should also have an idea of the style and color or finish of hardware you'd like.

Lastly, you should make a note of whether you'd like to include screens with your new windows. If so, you will likely have the option to choose between at least two options which differ in terms of visibility and durability, depending on the window brand you choose.

Ultimately, each window brand will have their own language to describe these options, and it can be difficult to make comparisons between them. If you've ever tried to compare mattresses made by different companies, it's a similar challenge.

Once you settle on a window brand that meets your other needs, you can evaluate their options on finishes and screens and ensure that they meet your needs.

Window Installation

This might be the most important chapter in this guide. You're going to entrust your home, which is probably your largest investment, to a contractor who'll ultimately make or break the whole project.

We've spoken with thousands of homeowners around the country, so we've seen the good, the bad, and the ugly of our industry. Everything we know about window installation can be summed up simply:

Even the best window will fail if installed improperly.

Watch the video in this post to see an example of this.



Window Installation

There are two reasons that people tend to hire the wrong window installers: Price and Time.



Replacing windows is a home improvement project that **can cost you a ton of money, headache, and damage to your home if done improperly**. As a general rule, anyone who promises that they can do it cheaper and faster may not be following all of the installation guidelines that guarantee a properly functioning window for years to come.

We can share countless examples of ways that contractors try to install windows for less money that wind up costing homeowners more over time.

One is using standard construction grade caulk. Our North Carolina humidity and temperature swings cause construction materials to expand and contract. In these conditions, the wrong caulk will become brittle and prone to water infiltration.

By contrast, we pay four times as much for a silicone based expansion caulk. This caulk will endure no matter what the weather does.*



IGNORE THE HYPE!

FIVE "MONEY SAVING TIPS"
FOR BUYING NEW WINDOWS
THAT WILL COST YOU MONEY

W

[Read this post for more detailed examples of money-saving tips that will ultimately cost you more.](#)

*Eventually all caulk will need to be replaced as an aspect of regular maintenance.

Window Installation

We recommend interviewing potential installers and asking them some or all of these questions.

- ***How long have you been in business?***

- You want to work with a professional who has a long history in your community. This is partly because you want to be able to see examples of their work and talk to other homeowners they've worked with. This also ensures that they have experience with North Carolina homes; our NC clay soil and weather are unique, and can pose some construction challenges. You want to work with someone who understands how homes settle, because this affects your window installation.

- ***How many window installation jobs did you do last year?***

- Ideally, you'll find a contractor who has plenty of experience installing windows, and does it frequently. **Since 2021, we've averaged 90 window installation jobs per year, with each job having a minimum of 3 windows installed.**

- ***Can I have a list of references?***

- An honest construction professional will be happy to provide references.

- ***Tell me about an installation job when you made a mistake.***

- Likewise, someone with lots of construction experience is bound to have made a mistake or two over the years. Look for someone who doesn't try to hide this fact. Instead, try to find someone who can speak honestly about what they did wrong, and, more importantly, what they did to make it right.

Window Installation

- ***What do you do that's different from what other installers do?***
 - Whether they talk about the materials they use, how they schedule jobs, or the professionalism of their team, listen for an answer that shows that this person understands the window installation business, and that they understand what sets them apart.
- ***What else do you do besides window installations?***
 - When it comes to choosing a window installer, you can choose someone who does a little bit of everything, or someone who just focuses on windows and doors. We are of course biased in our answer, because we only install windows and doors. If you do decide to contract a professional who does lots of home improvement projects, be sure that their references are for customers they've done window installation for.

Once you've finished your interview, ask yourself:

- How confident am I that this professional has the skills and professionalism needed to complete this major home renovation project?
- If there was a problem after the installation, how easy would it be for me to talk directly to the person who installed my windows for troubleshooting and repair if need be?

Window Installation

In addition to an interview, you should follow up by reading a prospective window installer's business reviews on Google.

[Here's a link to check out Google reviews for Window Works Co.](#)



What About Angie's List or Yelp?

You may be wondering, "Why check Google when there are home improvement review sites like Angie's List, or other review sites like Yelp?"

You should know that, unlike Google, these sites are generally 'pay to play'. Home improvement professionals who don't pay to advertise with these sites generally won't have all of their positive reviews visible to site visitors.

Only when you pay for ad space will they 'unlock' your full review history to page visitors.

We are fortunate at Window Works to have the benefit of positive word of mouth referrals from our customers. We don't pay for advertising on those sites, because we don't need to. But, that means that if you look us up there, you probably won't see all of our positive reviews.

Get in Touch!

Ready to start your window replacement project? Click here or on the calendar below to schedule a free, no obligation Window Consultation Call today!



☎ (919) 854-6121

✉ info@windowworksnc.com

🏠 8111 Tryon Woods Drive,
Suite 205, Cary, NC, 27518

🌐 www.windowworksnc.com